INCH-POUND

MIL-S-52300D
25 September 1991
SUPERSEDING
MIL-S-52300C
21 September 1983

MILITARY SPECIFICATION

SHOE ASSEMBLY, TRESTLE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a shoe assembly and its components to be used with a trestle assembly (MIL-T-52150).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir, Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5420

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SPECIFICATIONS

FEDERAL

PPP-B-601	-	Boxes, Wood, Cleated-Plywood.
PPP-B-621	-	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	-	Box, Shipping and Fiberboard.

MILITARY

MIL-T-704	-	Treatment and Painting of Materiel.

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.
MIL-STD-889	-	Dissimilar Metals.
MIL-STD-1186	-	Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; With Appropriate Test Methods.
MIL-STD-1190	-	Minimum Guidelines for Level C Preservation, Packing and Marking.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Document Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Government drawings. The following Government drawings form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those in effect on the date of the solicitation.

DRAWINGS

ME

TA13207E9716	-	Shoe Assembly, Trestle.

(Copies of drawings required by contractors in connection with specific acquisition functions should be obtained from the USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-JBS, Fort Belvoir, VA 22060-5606.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

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AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME - Boiler and Pressure Vessel Code, Section IX, Qualification Standard for Welding and Brazing Procedures, Welders, Brazers, and Welding and Brazing Operators.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3953 - Strapping, Flat Steel and Seals.
D 4675 - Selection and Use of Flat Strapping Materials.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN WELDING SOCIETY (AWS)

AWS A5.1 - Covered Carbon Steel Arc Welding Electrodes.
AWS A5.18 - Carbon Steel Filler Metals for Gas Shielded Arc Welding.
AWS D1.1 - Structural Welding Code - Steel.

(Application for copies should be addressed to the American Welding Society, 550 N.W LeJeune Road, P.O. Box 351040, Miami, FL 33135.)

(Non-Government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services).

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The shoe assembly and components shall be as shown on Top Assembly TA13207E9716 and as specified herein.

3.1.1 Drawings. The drawings forming a part of this specification are engineering design drawings. The contractor is responsible for preparing his own shop drawings. Where tolerances prescribed could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, and operation of the shoe assembly or components. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer.

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3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.3 Material. Material shall be as specified herein and as shown on the applicable drawing. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification. Material shall be new and unused.

3.3.1 Material deterioration prevention and control. The shoe assembly shall be fabricated from compatible material, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operating and storage environment to which the item may be exposed.

3.3.1.1 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.1.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with components and sub-components, and shall make information available upon request to the contracting officer or designated representative.

3.4 Welding electrodes. Welding electrodes shall conform to AWS A5.1, classification E60 or E70 series, or AWS A5.18, classification ER70S series, size as required.

3.5 Identification marking. The shoe assembly and components shall be metal die stamped as shown on the applicable drawing. Die stamping shall be legible after painting.

3.6 Treatment and painting. Specify the topcoat color for the shoe assembly and components, with the exception of the U-bolt assembly and the column shoe connecting bolt assembly which shall be cleaned, treated, and painted in accordance with MIL-T-704, type B.

3.7 Stenciling. The nomenclature, National stock number, weight, and cube shall be stenciled on each shoe assembly and components. The letters "NET WT" shall precede the numerals in the weight and letters "CU" shall precede the cubic displacement. Markings shall be gothic-type capitals and arabic numerals 1/2-inch high. Stenciling shall conform to MIL-T-704.

3.8 Workmanship.

3.8.1 Steel fabrication. Steel may be cut by shearing, sawing, or flame cutting. Burned surfaces of flame-cut material shall be ground or machined to remove cooling checks. All burrs and projections shall be removed.

3.8.2 Welders and welding.

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3.8.2.1 Welders and welding operators. Before assigning any welder or welding operator to welding work covered by this specification, the contractor shall provide the contracting officer with certification that the welder or welding operator has passed qualification tests as prescribed by either of the following listed codes for the materials joined and the type of welding operations to be performed and that such qualification is effective as defined by the particular code:

ASME Boiler and Pressure Vessel Code, Section IX, Qualification Standard for Welding and Brazing Procedures, Welders, Brazers, and Welding and Brazing Operators.

AWS D1.1, Structural Welding Code - Steel.

Contractors who make only horizontal welds need not qualify welders for "all position welding". In the event of evidence of poor welds, the Government reserves the right to require recertification of any welder or welding operator. The test results shall be made available for review by the contracting officer or the contracting officer's representative.

3.8.2.2 Welding. The surfaces of parts to be welded shall be free from scale, paint, grease, and other foreign matter. Welding shall be done by the shielded metal-arc or gas metal-arc (MIG) process and shall conform to the provisions of either ASME or AWS Code. Where embrittlement of the joint metal may result, preheating or annealing or both shall be employed. All welded parts shall be free from cracks and other imperfections that may reduce the effectiveness of the part. Work shall be positioned for flat welding when practicable. Before welding over previously deposited weld metal, all traces of slag shall be removed and the deposit and adjoining base metal shall be wire brushed clean.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance

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to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Parts and components. Parts and components detailed on the drawings shall be inspected in accordance with the quality assurance provisions (QAP) shown on the drawings. The drawings specify the characteristics requiring QAP inspection, the sampling plan, and the basis for acceptance and rejection (see 6.4 and 6.5).

4.2 Classification of inspections. Inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 Examination. The first article shoe assembly or components, as applicable, shall be examined for the defects specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Samples selected for examination shall be in accordance with MIL-STD-105 table I and table IIa. A lot shall be accepted when zero defects are found and rejected when one or more defects are found.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined for the major and minor defects as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.5 Inspection procedure.

4.5.1 Examination. The shoe assembly or components, as applicable, shall be examined as specified herein for the following defects:

Major

101. Materials not as specified (see 3.3).
102. Materials are not corrosion resistant or treated to be made corrosion resistant for the applicable storage and/or operating environment (see 3.3.1).
103. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other (see 3.3.1.1).
104. Contractor does not have documentation available for identification of material, material finishes or treatments (see 3.3.1.2).
105. Components or parts missing or not as specified (see 3.1).
106. Workmanship not as specified (see 3.8).

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107. Weld defects such as flux inclusions, spatter, undercutting, porosity, incorrect size, or lack of fusion with parent metal (see 3.8.2.2).
108. Castings containing blowholes, porosity, hard spots, shrinkage defects, cracks, or not smooth and well cleaned. (Minor imperfections not affecting strength may be corrected by welding (see 3.8.2.2).
109. Any part or component not in accordance with the QAP requirements as shown on the drawings (see 3.1.1).
110. Welding not as specified (see 3.8.2.2).
111. Welders and welding operators not certified as specified (see 3.8.2.1).

Minor

201. Dimensions other than those identified by QAP data not as specified (see 3.1.1).
202. Identification marking incorrect, missing, or illegible (see 3.5).
203. Treatment and painting not as specified (see 3.6).
204. Stenciling not as specified (see 3.7).

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105. Sample size shall be determined by using MIL-STD-105 table I and table IIa. A lot shall be accepted when zero defects are found and rejected when one or more defects are found.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. Presence of one or more defects shall be cause for rejection.

112. Containers not as specified for level A or B. Each incorrect container shall constitute one defect (see 5.1.1 and 5.1.2).
113. Boxes for shoe assemblies not provided with skids as specified for level A or B (see 5.1.1 and 5.1.2).
114. Components not secured within the boxes to prevent free movement as specified for level A or B (see 5.1.1, 5.1.1.1, 5.1.2 and 5.1.2.1).
115. Strapping not as specified for level A (see 5.1.1 and 5.1.1.1).
116. Gross weight or size exceeds the limitations of the fiberboard box for level B (see 5.1.2.1).
117. Fiberboard box not sealed and reinforced as specified for level B (see 5.1.2.1).
118. Packing not in accordance with the referenced document for level C (see 5.1.3).

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119. Marking missing, illegible, incorrect or incomplete for level A, B, or C (see 5.2).

5. PACKAGING

5.1 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.1.1 Level A. The components comprising two complete shoe assemblies shall be packed together in a close-fitting box conforming to PPP-B-621, class 2, style 2. The box shall be modified by the installation of skids in accordance with the box specification. Blocking and bracing to prevent movement and damage shall be provided and shall be in accordance with MIL-STD-1186. Box closure shall be in accordance with the appendix to the box specification. Strapping shall conform to ASTM D 3953, type 1 or 2, grade 1 or 2 and ASTM D 4675.

5.1.1.1 Bolt assemblies and U-bolt assemblies. When procured separately from the shoe assembly, assemblies of like description shall be packed together in close-fitting boxes conforming to PPP-B-621, class 2, style optional; or PPP-B-601, overseas type, style optional. The contents shall be secured within the box in a manner to prevent free movement in accordance with MIL-STD-1186. The boxes shall be closed in accordance with the appendix to the applicable box specification. Strapping shall conform to ASTM D 3953, type 1 or 2, grade 1 or 2 and ASTM D 4675.

5.1.2 Level B. The components comprising a complete shoe assembly shall be packed as specified for level A, except the box shall be class 1. Strapping may be zinc-coated.

5.1.2.1 Bolt assemblies and U-bolt assemblies. Assemblies of like description procured separately from the shoe assembly shall be packed in close-fitting boxes conforming to PPP-B-621, class 1, style optional; PPP-B-601, domestic type, style optional; or PPP-B-636, V3c, style optional. The contents shall be secured within the box in a manner to prevent free movement in accordance with MIL-STD-1186. The gross weight or size of fiberboard boxes shall not exceed the limitations of the box specification, and the boxes shall be waterproof sealed and reinforced with bands of reinforced tape in accordance with the appendix to the box specification. Strapping on wood boxes may be zinc-coated.

5.1.3 Level C. The components comprising a complete shoe assembly or the bolt assembly and U-bolt assembly procured separately from the shoe assembly, shall be packed in accordance with MIL-STD-1190.

5.2 Marking. In addition to any special marking specified in the contract or purchase order (see 6.2), marking shall be in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

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6.1 Intended use. The shoe assembly and its components are intended for use in floating and fixed bridges as a spread footing and bearing socket for the column of a trestle assembly. This item is a component of the following Supply Catalogs (SC), SC 5420-97-E03, E35, E36 and E46.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. When a first article is required for inspection and approval and the number of units required (see 3.2).
d. Level of packing required (see 5.1).
e. Any special marking (see 5.2).

6.3 First article. When a first article inspection is required, the item(s) should be a preproduction model. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of the first article test results and disposition of the first articles. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Quality assurance provisions (QAP) records. The contracting officer should require the contractor to maintain records of all QAP inspections. A suggested paragraph is as follows:

"The contractor shall maintain complete records of all examinations and tests performed to verify the requirements of classified QAP characteristics. The records shall include as a minimum, lot size, sample size, drawing requirement, actual measurement, number and type of deficiencies found, quantity approved, quantity rejected, corrected action taken when applicable."

6.5 Definition.

6.5.1 Quality assurance provisions (QAP). A QAP is a contractual requirement that supplements section 4 of the specification. QAP indicates the minimum requirements which must be inspected on the product drawings to verify the design objectives of the product and assure interchangeability of repair parts.

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes which respect to the previous issue due to the extensiveness of the changes.

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6.7 Subject term key word listing.

Bridge
Shoe Assembly
Trestle

Custodians:
Army - ME

User activity:
Navy - MC

Preparing activity:
Army - ME

Project 5420-0181

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
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RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-S-52300D	2. DOCUMENT DATE (YYMMDD) 910925

3. DOCUMENT TITLE
Shoe Assembly, Trestle

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (if applicable) (2) AUTOVON	7. DATE SUBMITTED

8. PREPARING ACTIVITY

a. NAME Gerry Harlow	b. TELEPHONE (Include Area Code) (1) Commercial (703) 664-1806 (2) AUTOVON 354-1806
c. ADDRESS (Include Zip Code US Army Belvoir RDE Center ATTN: STRBE-TSE Fort Belvoir, VA 22060-5606	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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